Discrete Mathematics and Statistics (CPT107)

1: If X and Y are empty sets, then X = Y. If you think that it is true, prove it. If not, explain why.

2: Let X, Y and Z be sets. If $\mathrm{X} \subset \mathrm{Y}$ and $\mathrm{Y} \subset \mathrm{Z}$, then $\mathrm{X} \subset \mathrm{Z}$. If you think that it is true, prove it. If not, explain why.

3: Let X, Y and Z be sets. Is this expression $(X-Y)\cup(Y-Z)=X-Z$ true? If your answer is yes, then prove it. Otherwise, give a counterexample.

4: Prove that the following sets are equal:

1. $A \times (B \cup C) = (A \times B) \cup (A \times C)$
2. $A \times (B \cap C) = (A \times B) \cap (A \times C)$
3. $A \times (B - C) = (A \times B) - (A \times C)$

5: Let A and B be two sets. Prove the following statements: $A \subseteq B$ if and only if $\sim B \subseteq \sim A$.